

Session 3 Homework Solutions

APSTA-GE 2006: Applied Statistics for Social Science Research

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This document provides worked solutions to the [Lecture 3 homework](#).

1 Inspecting the iris data

Task: Take a look at the `iris` dataset using `?iris`.

The built-in `iris` data frame contains 150 observations and five variables. Four variables are numeric flower measurements in centimeters, and `Species` identifies one of three iris species. Each species contributes 50 observations.

```
dim(iris)
```

```
[1] 150  5
```

```
names(iris)
```

```
[1] "Sepal.Length" "Sepal.Width"  "Petal.Length" "Petal.Width"  "Species"
```

```
table(iris$Species)
```

```
      setosa versicolor virginica  
        50         50         50
```

```
head(iris)
```

```
      Sepal.Length Sepal.Width Petal.Length Petal.Width Species  
1           5.1         3.5         1.4         0.2   setosa  
2           4.9         3.0         1.4         0.2   setosa  
3           4.7         3.2         1.3         0.2   setosa  
4           4.6         3.1         1.5         0.2   setosa  
5           5.0         3.6         1.4         0.2   setosa  
6           5.4         3.9         1.7         0.4   setosa
```

2 Overall means

Task: Compute the overall means of `Sepal.Length`, `Sepal.Width`, `Petal.Length`, and `Petal.Width`.

Because the four measurement columns are numeric, `across(where(is.numeric), mean)` applies `mean()` to all four in one step.

```
overall_means <- iris |>  
  summarise(across(where(is.numeric), mean))  
  
knitr::kable(  
  overall_means,  
  digits = 3,  
  caption = "Overall means of the iris measurements"  
)
```

Table 1: Overall means of the iris measurements

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
5.843	3.057	3.758	1.199

Thus, the overall means are approximately 5.843 for sepal length, 3.057 for sepal width, 3.758 for petal length, and 1.199 for petal width.

3 Means by species

Task: Compute the four means by `Species` using `group_by()` and `summarise()`.

`group_by(Species)` partitions the rows by species. The following `summarise()` call then computes each mean separately within those groups.

```
species_means <- iris |>
  group_by(Species) |>
  summarise(
    across(where(is.numeric), mean),
    .groups = "drop"
  )

knitr::kable(
  species_means,
  digits = 3,
  caption = "Mean iris measurements by species"
)
```

Table 2: Mean iris measurements by species

Species	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
setosa	5.006	3.428	1.462	0.246
versicolor	5.936	2.770	4.260	1.326
virginica	6.588	2.974	5.552	2.026

The grouped means reveal substantial differences in petal size: setosa has much shorter and narrower petals than versicolor and virginica.

4 Petal-length and sepal-length scatter plot

Task: Reproduce the scatter plot of `Sepal.Length` against `Petal.Length`, colored by `Species`.

Each point represents one flower. Mapping `Species` to color makes the three groups visible, while the axes reproduce the reference plot.

```
scatter_plot <- ggplot(
  iris,
  aes(x = Petal.Length, y = Sepal.Length, color = Species)
) +
  geom_point(size = 2) +
  labs(
    x = "Petal.Length",
```

```

    y = "Sepal.Length",
    color = "Species"
  )
scatter_plot

```

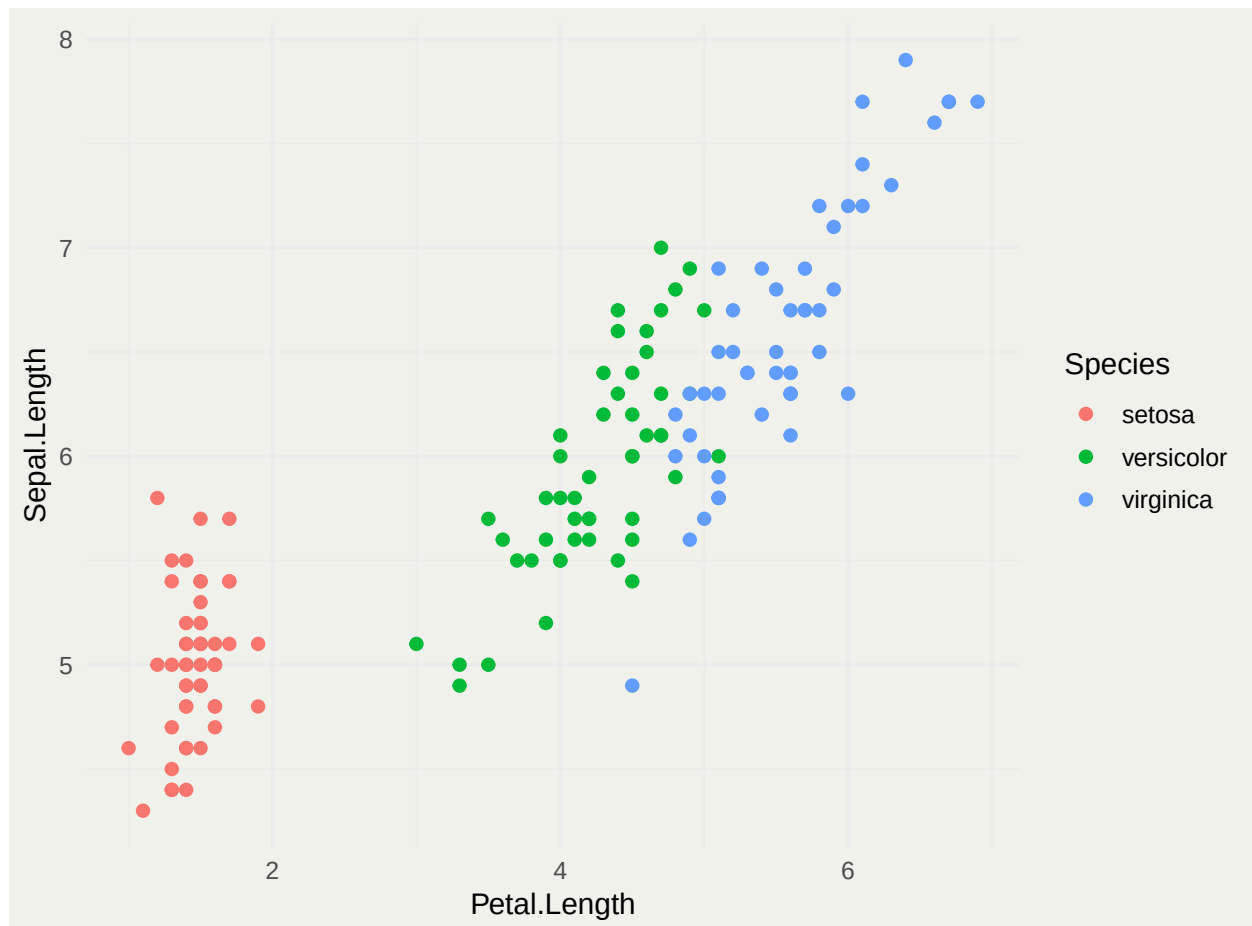


Figure 1: Sepal length versus petal length, colored by species.

The plot shows clear species clustering and a positive relationship between petal length and sepal length.

5 Faceted density plots

Task: Reproduce the faceted density plot using `geom_density()` and `facet_wrap()`.

The measurements must first be transformed from wide to long form. In the long data, `name` identifies the measurement and `value` stores its observed value. We then draw one density panel per measurement.

```
iris_long <- iris |>
  pivot_longer(
    cols = -Species,
    names_to = "name",
    values_to = "value"
  )

density_plot <- ggplot(
  iris_long,
  aes(x = value, color = Species, fill = Species)
) +
  geom_density(alpha = 1/5) +
  facet_wrap(~ name, scales = "free", ncol = 2) +
  labs(x = NULL, y = "density", color = "Species", fill = "Species")

density_plot
```

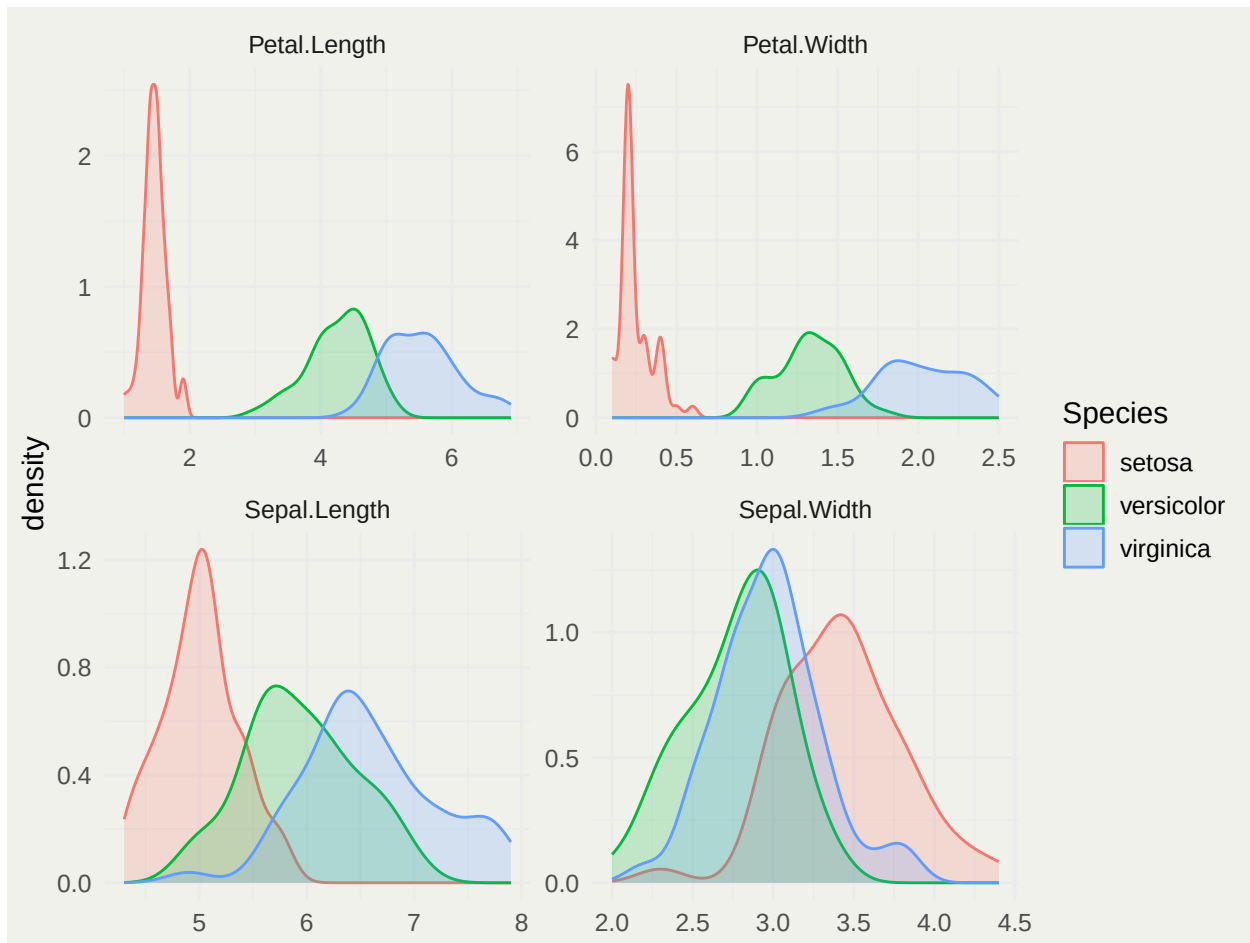


Figure 2: Distributions of the four iris measurements by species.

Because `name` is a character column, `facet_wrap()` orders the labels alphabetically by default,

which matches the reference plot. The free scales are important because petal width and sepal length occupy very different numerical ranges.

6 Indefinite integral

Task: Compute the following integral and show the steps:

$$\int 2x \cos(x^2) dx.$$

Use substitution. Let

$$u = x^2.$$

Then

$$du = 2x dx,$$

so the integral becomes

$$\begin{aligned} \int 2x \cos(x^2) dx &= \int \cos(u) du \\ &= \sin(u) + C \\ &= \boxed{\sin(x^2) + C}. \end{aligned}$$

Differentiating the result checks the calculation:

$$\frac{d}{dx} \sin(x^2) = \cos(x^2)(2x) = 2x \cos(x^2).$$

7 Definite integral from 0 to 2π

Task: Evaluate the integral on paper from 0 to 2π , then validate the result with R's `integrate()` function.

Using the antiderivative above,

$$\begin{aligned} \int_0^{2\pi} 2x \cos(x^2) dx &= [\sin(x^2)]_0^{2\pi} \\ &= \sin((2\pi)^2) - \sin(0) \\ &= \boxed{\sin(4\pi^2)} \\ &\approx 0.9783406. \end{aligned}$$

Notice that $\sin(4\pi^2)$ is not zero: the argument is $4\pi^2$, not an integer multiple of 2π .

```
integrand <- function(x) {  
  2 * x * cos(x^2)  
}  
  
numeric_result <- integrate(  
  integrand,  
  lower = 0,  
  upper = 2 * pi  
)  
  
numeric_result
```

0.9783406 with absolute error < 4.1e-11

```
c(  
  analytic = sin(4 * pi^2),  
  numerical = numeric_result$value,  
  difference = abs(sin(4 * pi^2) - numeric_result$value)  
)
```

analytic	numerical	difference
9.783406e-01	9.783406e-01	1.687539e-14

The analytical and numerical values agree to numerical precision.